Notes: Ben-Rephael, Da & Israelsen (RFS, 2017)

It Depends on Where You Search: Institutional Investor Attention and Underreaction to News

Contents

1 Big picture

- **Question.** Does investor attention affect how quickly stock prices incorporate news, and does the answer depend on whether attention comes from institutional or retail investors?
- **Main innovation.** The paper constructs a direct measure of abnormal institutional investor attention using Bloomberg terminal news-searching and news-reading activity for individual stocks.
- **Core measure.** Bloomberg assigns each stock-day a score from 0 to 4 based on abnormal Bloomberg terminal attention relative to the prior 30 days. The authors define AIA = 1 when the score is 3 or 4.
- **Why Bloomberg matters.** Bloomberg terminals are used primarily by professional investors. In the authors' user-profile evidence, the largest job categories are portfolio/fund/investment managers, analysts, and traders; the largest industries are banking, asset management, and institutional financial services.
- **Contrast with retail attention.** Retail attention is measured with Google search activity, following Da, Engelberg, and Gao's abnormal search volume index approach.
- **First main result.** Institutional attention is related to news, returns, volatility, institutional trading volume, retail attention, and EDGAR search activity, but it is not the same as these other attention proxies.
- **Second main result.** Institutional attention responds faster to firm news and leads retail attention. A shock to institutional attention predicts future retail attention, while the reverse relation is much weaker.
- **Third main result.** Institutional attention facilitates permanent price adjustment. If institutions pay attention to an earnings announcement or analyst recommendation change, the announcement-day price response is stronger and subsequent drift is much weaker.
- **Key asset-pricing implication.** Post-earnings-announcement drift and recommendation-change drift are concentrated in events with AIA = 0, meaning events ignored by institutions.
- **Trading implication.** Long-short strategies based on earnings surprises or analyst recommendation changes earn large returns when they trade the AIA = 0 events, but the analogous AIA = 1 strategies are much closer to zero.
- **Economic takeaway.** Limited attention matters most when the investors capable of moving prices fail to process the information. Retail attention alone is not enough to explain permanent information incorporation.

2 Data and measurement

2.1 Sample construction

- The sample covers Russell 3000 common stocks with CRSP share codes 10 and 11.
- The time period is February 2010 to December 2015.
- The full stock-day sample contains 3,144,109 observations.
- The earnings-announcement sample contains 34,400 firm-quarter observations.
- The analyst recommendation-change sample contains 16,312 recommendation changes.
- Standard firm data are drawn from CRSP, Compustat, I/B/E/S, RavenPack/Dow Jones news, Google SVI, EDGAR logs, and Ancerno institutional trading data.

Interpretation: The sample is broad enough to study daily stock-level attention and returns, while the event samples isolate two classic sources of underreaction: earnings news and analyst recommendation revisions.

2.2 Abnormal institutional attention

Bloomberg records the number of times terminal users read articles about a stock and the number of times they search for news on that stock. Bloomberg then compares each hourly count to the stock's previous 30-day distribution.

The paper's main institutional-attention dummy is

$$\mathrm{AIA}_{i,t} = \mathbf{1}\left\{\text{Bloomberg score}_{i,t} \in \{3,4\}\right\}. \tag{1}$$

The Bloomberg score takes values 0, 1, 2, 3, or 4. The authors also construct a continuous version, AIAC, by mapping Bloomberg scores to conditional means of a truncated normal:

$$\text{score } 0,1,2,3,4 \quad \mapsto \quad -0.35,\ 1.045,\ 1.409,\ 1.647,\ 2.154. \tag{2}$$

Interpretation: AIA identifies the right tail of institutional search and reading activity. It is deliberately event-like: it asks whether institutional investors suddenly pay unusual attention to a stock on a given day.

2.3 Retail attention

Retail attention is measured using Google Search Volume Index data. The abnormal retail attention dummy is constructed analogously:

$$\mathrm{DADSVI}_{i,t} = \mathbf{1}\left\{\text{Google-based abnormal search score}_{i,t} \in \{3,4\}\right\}. \tag{3}$$

The continuous Google attention variable is denoted ADSVI.

Interpretation: Google attention is more likely to capture retail search behavior, while Bloomberg terminal attention is more likely to capture institutional search and reading behavior. The paper's central design is to compare these two direct attention measures.

2.4 Other attention and information variables

- $\mathrm{ANews}_{i,t}$: log ratio of one plus the number of Dow Jones newswire articles about the firm on day t to the prior-month average.
- $EDGAR_{i,t}$: unique SEC EDGAR filing requests for the firm.
- $\mathrm{AEDGAR}_{i,t}$: log ratio of EDGAR requests to the previous month's average.
- $AVol_{i,t}$: abnormal total trading volume.
- $Ancerno\text{-}AVol_{i,t}$: abnormal institutional trading volume using Ancerno institutional trade data.
- $AbsRet$, $AbsDGTW$, $HLtoH$, $52HighDum$, $52LowDum$: price-movement and salience controls.

Interpretation: These variables separate institutional attention from other common proxies for information flow, salience, and trading intensity.

2.5 Event variables

- SUE: quarterly standardized unexpected earnings from I/B/E/S, using the actual earnings minus the most recent analyst forecast, divided by the forecast standard deviation.
- RecChng: analyst recommendation change, ranging from -4 to $+4$, where positive values indicate upgrades and negative values indicate downgrades.
- $DGTW$: Daniel, Grinblatt, Titman, and Wermers risk-adjusted returns.

Interpretation: Earnings surprises and recommendation changes provide clean settings in which the sign and magnitude of public information can be measured directly.

3 Models and empirical specifications

3.1 Determinants of attention

The first set of regressions studies what predicts institutional and retail attention. The generic probit specification is

$$\Pr\left(\mathrm{AIA}_{i,t}=1\right)=\Phi\left(\alpha+\beta' X_{i,t}+\varepsilon_{i,t}\right), \tag{4}$$

where $X_{i,t}$ includes news, earnings-announcement and recommendation-change dummies, price and volume variables, firm characteristics, other attention measures, and day-of-week controls.

The analogous retail specification is

$$\Pr\left(\mathrm{DADSVI}_{i,t}=1\right)=\Phi\left(\alpha+\gamma' X_{i,t}+u_{i,t}\right). \tag{5}$$

Interpretation: These regressions ask whether institutional attention is merely another label for news, volatility, trading volume, retail search, or EDGAR activity. The answer is no: it is related to all of them but retains distinct variation.

3.2 Institutional attention and institutional trading

To link AIA with institutional trading, the paper estimates abnormal-volume regressions of the form

$$Y_{i,t}=\alpha+\beta \mathrm{AIA}_{i,t}+\Gamma' Controls_{i,t}+\varepsilon_{i,t}, \tag{6}$$

where $Y_{i,t}$ is either abnormal CRSP volume or abnormal Ancerno institutional volume.

It also compares the AIA coefficient with the retail-attention coefficient using a difference-in-differences style comparison between total abnormal volume and institutional abnormal volume.

Interpretation: If Bloomberg attention captures institutional attention, it should be especially related to institutional trading. Table 4 confirms this link.

3.3 Lead-lag relation among institutional attention, retail attention, and news

The paper estimates a panel VAR with five lags:

$$\mathrm{AIAC}_{i,t}=\alpha_1+\sum_{j=1}^{5}\beta_{1j}\mathrm{AIAC}_{i,t-j}+\sum_{j=1}^{5}\gamma_{1j}\mathrm{ADSVI}_{i,t-j}+\sum_{j=1}^{5}\delta_{1j}\mathrm{ANews}_{i,t-j}+\Gamma_1' Z_{i,t}+\mu_i+\varepsilon_{1,i,t}, \tag{7}$$

$$\mathrm{ADSVI}_{i,t}=\alpha_2+\sum_{j=1}^{5}\beta_{2j}\mathrm{AIAC}_{i,t-j}+\sum_{j=1}^{5}\gamma_{2j}\mathrm{ADSVI}_{i,t-j}+\sum_{j=1}^{5}\delta_{2j}\mathrm{ANews}_{i,t-j}+\Gamma_2' Z_{i,t}+\mu_i+\varepsilon_{2,i,t}, \tag{8}$$

$$\mathrm{ANews}_{i,t}=\alpha_3+\sum_{j=1}^{5}\beta_{3j}\mathrm{AIAC}_{i,t-j}+\sum_{j=1}^{5}\gamma_{3j}\mathrm{ADSVI}_{i,t-j}+\sum_{j=1}^{5}\delta_{3j}\mathrm{ANews}_{i,t-j}+\Gamma_3' Z_{i,t}+\mu_i+\varepsilon_{3,i,t}. \tag{9}$$

Interpretation: The VAR is designed to ask who reacts first. Institutional attention responds quickly to news and predicts retail attention; retail attention has weaker ability to predict institutional attention.

3.4 Earnings-announcement return regressions

For earnings announcements, the paper estimates return regressions at announcement day and future horizons:

$$R^h_{i,t} = \alpha + \beta_1 \mathrm{AIA}_{i,t} + \beta_2 \mathrm{SUE}_{i,t} + \beta_3 (\mathrm{SUE}_{i,t} \times \mathrm{AIA}_{i,t}) + \Gamma' Controls_{i,t} + FE + \varepsilon^h_{i,t}. \tag{10}$$

The horizon h is either the announcement-day return or cumulative DGTW-adjusted returns from $t+1$ to $t+j$, with $j = 1, 2, 3, 5, 10, 20, 30, 40$.

Interpretation: A positive β_3 on announcement day means institutional attention strengthens immediate price reaction. A negative β_3 in future cumulative returns means institutional attention reduces post-announcement drift.

3.5 Analyst recommendation-change return regressions

For analyst recommendation changes, the paper estimates

$$R^h_{i,t} = \alpha + \theta_1 \mathrm{AIA}_{i,t} + \theta_2 \mathrm{RecChng}_{i,t} + \theta_3 (\mathrm{RecChng}_{i,t} \times \mathrm{AIA}_{i,t}) + \Gamma' Controls_{i,t} + FE + \varepsilon^h_{i,t}. \tag{11}$$

Interpretation: The logic is identical to the earnings-announcement test. Institutional attention should make the immediate price reaction larger and the subsequent recommendation drift smaller.

3.6 Calendar-time trading strategies

The paper constructs long-short strategies separately for AIA = 0 and AIA = 1 events. For earnings announcements, the basic strategies are

$$LS\_AIA0 = R(\mathrm{AIA} = 0, \mathrm{SUE} > 0) - R(\mathrm{AIA} = 0, \mathrm{SUE} < 0), \tag{12}$$

$$LS\_AIA1 = R(\mathrm{AIA} = 1, \mathrm{SUE} > 0) - R(\mathrm{AIA} = 1, \mathrm{SUE} < 0), \tag{13}$$

$$DIFF = LS\_AIA0 - LS\_AIA1. \tag{14}$$

For analyst recommendation changes, SUE is replaced by RecChng.

Interpretation: These strategies translate the regression evidence into tradable calendar-time return patterns. If drift comes mainly from inattentive institutions, LS_AIA0 should earn larger returns than LS_AIA1.

4 Identification and empirical design

4.1 Why the attention measure is useful

- It is direct: it comes from actual news search and reading activity rather than from equilibrium outcomes such as returns or volume.
- It is stock-specific and daily, making it suitable for event studies.
- It separates institutional attention from retail attention by using Bloomberg terminals instead of Google searches.
- It distinguishes “news exists” from “professional investors actually look at the news.”

Interpretation: The paper's identification strength comes from measuring the missing cognitive input directly: whether institutions pay attention.

4.2 What variation identifies the price-response results

- Earnings surprises and recommendation changes provide measurable information shocks.
- Conditional on the same event type and sign, the paper compares events with AIA = 1 versus AIA = 0.
- Controls absorb other attention proxies, news intensity, trading volume, volatility, salience, size, book-to-market, analyst coverage, institutional holdings, spreads, and prior returns.
- Quarter and day-of-week fixed effects are included in the event-return regressions.

Interpretation: The key identifying comparison is not whether news moves prices. It is whether similar public news moves prices faster when institutions pay attention.

4.3 What makes the lead-lag evidence informative

- The VAR includes lags of institutional attention, retail attention, and news.
- It uses firm fixed effects and controls for contemporaneous exogenous variables.
- The impulse responses show that institutional attention reacts more quickly to news and leads retail search.

Interpretation: This supports the paper's interpretation that institutions are often the first attentive traders after news, while retail investors respond more slowly.

4.4 Limitations

- Bloomberg terminal data are transformed scores, not raw read and search counts.
- AIA is not randomly assigned. News that draws institutional attention may differ in unobserved ways from news that does not.
- The measure captures Bloomberg users, not all institutions.
- The paper studies two major event types; other corporate events may have different attention dynamics.

Interpretation: The evidence is best read as strong, carefully controlled evidence that institutional attention is central to information incorporation, rather than as a randomized experiment.

5 Tables and figures

This section keeps the same *Read* plus *Economic message* format. Adjacent visuals are grouped to save space and improve comparison. The visual panels are cropped directly from the original paper.

5.1 Figure 1 and Figure 2: What the attention measure looks like and who uses Bloomberg

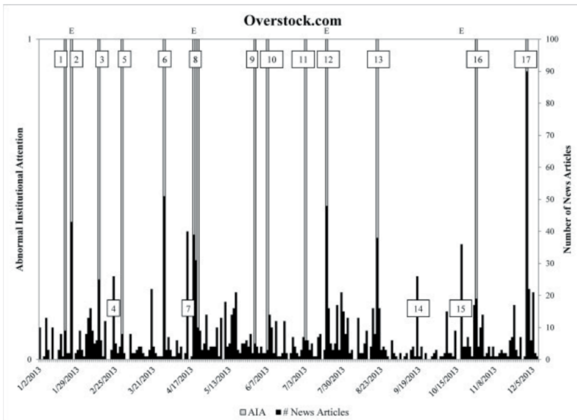
Read: Figure 1:

- Overstock.com experiences many AIA spikes during 2013.
- Some AIA spikes occur around earnings announcements, but many occur around other salient corporate news.
- News articles and institutional attention are correlated, but news does not mechanically guarantee AIA.

Read: Figure 2:

- The largest job-title categories are portfolio/fund/investment managers, analysts, and traders.
- The largest industry categories are banking, asset management, and institutional financial services.
- This supports the interpretation that Bloomberg terminal activity primarily reflects institutional investors.

Economic message: The paper's contribution starts with measurement: it observes professional investors searching and reading firm news, not just price or volume outcomes.



Event	Date	AIA	Sample Headline
1	1/18/2013	1	*Airport police: Overstock CEO arrested for having gun in luggage*
2	1/24/2013	1	*Overstock.com 4Q EPS 37c*
3	2/12/2013	1	*Overstock.com CEO to Take Medical Leave of Absence*
4	2/22/2013	0	*Overstock.com Inc Announces President Change-Form 8-K*
5	2/28/2013	1	*Groupon Shares Plunge on Profit, International Concerns*
6	3/28/2013	1	*Overstock.com Eyeing Supreme Court Appeal of Adverse NY Internet Tax Decision*
7	4/15/2013	0	*DJ Overstock CEO Byrne Resumes Duties After Medical Leave of Absence*
8	4/18/2013	1	*Overstock.com Reports Q1 2013 Results*
8	4/19/2013	1	*Overstock.com Raised to Buy From Underperform by BofA-Merrill Lynch*
9	5/30/2013	1	*Top 10 Nasdaq-traded stocks posting largest percentage decreases*
10	6/7/2013	1	*Officer JOHNSON III Sells 2,000 Of OVERSTOCK.COM INC*
11	7/3/2013	1	*Overstock.com downgraded at BofA/Merrill*
12	7/18/2013	1	*Overstock 2nd-quarter profit grows more than sevenfold, shares surge to multiyear high*
13	8/21/2013	1	*Overstock.com Victorious In Federal Lawsuit*
14	9/18/2013	0	*Overstock.com's 'The Good Good' sweepstakes awards entrants and their favorite charity*
15	10/17/2013	0	*Overstock.com's 3rd-quarter net income rises 31 percent as sales jump*
16	10/28/2013	1	*Overstock.com Presents Hot 99.5's Second Annual Jingle Ball in Washington, D.C.*
17	12/2/2013	1	*U.S. Supreme Court Won't Review New York Sales-Tax Law For Online Retailers*

(a) Institutional abnormal attention, earnings announcements, and news.

(b) Breakdown of Bloomberg terminal users.

5.2 Table 1 and Table 2: Variable definitions and summary statistics

Read: Table 1:

- AIA is a dummy equal to one when Bloomberg's transformed news-search/read score is 3 or 4.
- AIAC maps the 0–4 score into continuous conditional means of a truncated normal.
- The paper also defines Google attention, EDGAR attention, news, volume, volatility, salience, firm characteristics, and event variables.

Read: Table 2:

- In the full sample, average AIA frequency is 0.089, meaning the average stock experiences institutional attention shocks on about 8.9% of trading days.
- AIA frequency rises to 0.620 around earnings announcements and 0.478 around recommendation changes.
- Conditional on AIA = 1, stocks have larger absolute returns, larger abnormal volume, more EDGAR search, and higher retail search attention.

Economic message: Institutions do not pay abnormal attention randomly. Attention clusters around important events and high-salience trading days, but it remains distinct from news and retail search.

Table 1
Variable definitions

Variable	Definition
Bloomberg attention variables	
AIA	Bloomberg records the number of times news articles on a particular stock are read by its terminal users and the number of times users actively search for news for a specific stock. Bloomberg then assigns a value of one for each article read and ten for each news search. These numbers are then aggregated into an hourly count. Using the hourly count, Bloomberg then creates a numerical attention score each hour by comparing the past eight-hour average count to all hourly counts over the previous month for the same stock. They assign a value of zero if the rolling average is in the lowest 80% of the hourly counts over the previous 30 days. Similarly, Bloomberg assigns a score of 1, 2, 3, or 4 if the average is between 80% and 90%, 90% and 94%, 94% and 96%, or greater than 96% of the previous 30 days' hourly counts, respectively. Finally, Bloomberg aggregates up to the daily frequency by taking a maximum of all hourly scores throughout the day. These are the data provided to us by Bloomberg. Since we are interested in abnormal attention, our *AIA* measure is a dummy variable that receives a value of one if Bloomberg's score is 3 or 4, and zero otherwise. This captures the right tail of the measure's distribution
AIAC	We transform Bloomberg's 0, 1, 2, 3, and 4 scores to continuous values using the conditional means of the truncated normal distribution. The values are −0.35, 1.045, 1.409, 1.647, and 2.154, respectively. For example, 1.647 is the conditional mean of a standard normal random variable x, for x between NORMINV(0.94) and NORMINV(0.96), where NORMINV is the standard normal inverse cumulative distribution function
Other direct attention variables	
ADSVI	Da, Engelberg, and Gao's (2011) abnormal retail attention measure calculated as the natural log of the ratio of *DSVI* on day *t* to the average of *DSVI* over the previous month. *DSVI* is Google's daily Search Volume Index (*SVI*)
DADSVI	We follow Bloomberg's methodology and assign *DSVI* on day *t*one of the potential 0, 1, 2, 3, or 4 scores using the firm's past 30 trading day *DSVI* values. For example, if *DSVI* on day *t* is in the lowest 80% of past *DSVI* values, it receives the score zero. *DADSVI* is one on day *t* if the score is 3 or 4, and zero otherwise
EDGAR	The daily number of unique requests for firm filings on the SEC EDGAR server (Loughran and McDonald 2015). *EDGAR* is available until March 2015
AEDGAR	The natural log of the ratio of *EDGAR* on day *t* to the average of *EDGAR* over the previous month
Other variables	
ANews	The natural log of the ratio of one plus the number of news articles published on the Dow Jones newswire during the day to its average over the previous month. News data are provided by RavenPack
Ancerno-AVol	The stock's Ancerno daily volume divided by the previous eight-week average Ancerno trading volume. Ancerno data are available until June 2015
CRSP-AVol	The stock's total daily volume in CRSP divided by the previous eight-week average total trading volume
EarnDum	A dummy variable that is equal to one on earnings announcements days and zero otherwise
RecChngDum	A dummy variable that is equal to one on days with a change in analyst recommendations and zero otherwise
SUE	The quarterly standardized unexpected earnings calculated from I/B/E/S as the quarter's actual earnings minus the average of the most recent analyst forecast, divided by the standard deviation of that forecast
RecChng	The change in analyst recommendations. The variable ranges from −4 to 4, where a positive (negative) number refers to an upgrade (a downgrade)
Ret	CRSP's daily stock return
AbsRet	Absolute value of *Ret*
DGTW	CRSP's daily stock return minus the stock's benchmark portfolio daily return following Daniel et al. (1997)
AbsDGTW	Absolute value of *DGTW*
AVol	The stock's abnormal trading volume calculated following Barber and Odean (2008) as the stock's daily volume divided by the previous 252 day average trading volume

(a) Table 1, first part.

Table 1
Continued

Variable	Definition
HLtoH	The ratio between the stock's daily high and low price difference and the daily high price
52HighDum	A dummy variable that is equal to one if the stock's price exceeds its 52-week high price and zero otherwise
52LowDum	A dummy variable that is equal to one if the stock's price falls below its 52-week low price and zero otherwise
Turnover	The daily stock turnover
Dvol	The daily dollar trading volume in millions of dollars
Relative Spread	[(Ask-Bid)/Midpoint]/2 using *CRSP* end of day quotes
SizeInM	Stock's market capitalization, rebalanced every June, in millions of dollars
LnSize	The log of the stock's average size in millions of dollars from day *t-27* to *t-6*
LnBM	The natural logarithm of the firm's book-to-market ratio rebalanced every June following Fama-French (1992)
SDRET	The standard deviation of daily stock returns from day *t*-27 to day *t*-6
InstHold	The percentage of shares held by institutional investors obtained from the Thomson Reuters CDA/Spectrum institutional holdings' (S34) database
NumEst	The number of analysts covering the stock using the most recent information
LnNumEst	Log(1+*NumEst*)
AdvExpToSales	The firm's advertising expenses to sales as in Da, Engelberg, and Gao (2011) and using the most recent information
Tuesday – Friday	Dummy variables equal to one if the stock's day of the week is Tuesday-Friday, respectively, and zero otherwise

(b) Table 1, continuation.

Table 2
Summary statistics of *AIA* and other selected variables

A. Cross-sectional statistics

	Full sample			*EarnAnn* sample			*RecChng* sample		
Variables	Mean	Median	SD	Mean	Median	SD	Mean	Median	SD
Num firms	2,669			2,231			2,068		
AIA	0.089	0.070	0.082	0.620			0.478		
DADSVI	0.092	0.096	0.030	0.173			0.132		
SizeInM	6,171	1,120	21,529	6,511	1,211	22,460	7,258	1,609	23,007
BM	0.69	0.61	0.78	0.62	0.54	0.45	0.61	0.51	0.55
SDRET	2.25	2.05	0.96	2.10	1.87	1.04	2.24	1.97	1.22
Ret %	0.05	0.06	0.12	0.17	0.19	2.81	0.25	0.21	4.55
DGTW %	0.01	0.01	0.11	0.14	0.13	2.67	0.17	0.14	4.39
Turnover	0.01	0.01	0.01	0.03	0.02	0.03	0.02	0.02	0.03
Dvol	55.33	10.66	198.30	141.15	31.02	508.27	107.10	33.51	303.56
HLtoH	0.03	0.03	0.01	0.07	0.06	0.03	0.05	0.04	0.03
InstHold	0.60	0.62	0.20	0.59	0.64	0.19	0.63	0.67	0.19
NumEst	9.12	7.04	7.06	9.45	7.38	6.74	10.98	9.10	6.94
Abs SUE/REC	N/A	N/A	N/A	2.75	2.31	2.03	1.37	1.33	0.31
EDGAR	51.82	31.59	86.71	97.33	61.63	139.55	72.02	41.58	119.51
AEDGAR	−0.04	−0.02	0.07	0.56	0.56	0.33	0.15	0.12	0.41

B. Sample averages conditioning on AIA

	Full sample		*EarnAnn* sample		*RecChng* sample	
Variables	AIA=0	AIA=1	AIA=0	AIA=1	AIA=0	AIA=1
AbsRet %	1.55	3.29	4.47	5.85	2.75	4.67
AbsDGTW %	1.26	2.94	4.19	5.47	2.43	4.33
Turnover	0.009	0.020	0.019	0.031	0.019	0.032
Dvol	49.88	80.02	69.74	159.65	87.85	151.06
HLtoH	0.030	0.047	0.065	0.072	0.040	0.051
Relative spread	0.0010	0.0008	0.0010	0.0006	0.005	0.0004
NumEst	9.11	9.47	7.57	10.26	11.23	12.58
InstHold	0.60	0.59	0.58	0.62	0.64	0.65
Abs SUE/REC	n/a	n/a	2.74	2.87	1.37	1.38
DADSVI	0.086	0.145	0.130	0.191	0.112	0.154
EDGAR	49.73	67.84	69.62	105.19	67.34	85.83
AEDGAR	−0.073	0.206	0.470	0.627	0.081	0.220

The table reports the summary statistics of our Abnormal Institutional Attention measure (*AIA*) from Bloomberg and other selected variables from February 2010 to December 2015. Our initial sample includes all Russell 3000

Figure 3: Table 2: Summary statistics of AIA and selected variables.

5.3 Table 3: What drives institutional and retail attention?

Read:

- Panel A uses AIA as the dependent variable; Panel B uses DADSVI.
- News variables, earnings announcements, and recommendation-change dummies strongly predict both institutional and retail attention.
- Price movement, abnormal volume, high-low price range, 52-week high/low indicators, size, analyst coverage, and EDGAR activity also matter.
- ADSVI predicts AIA, and AIA predicts DADSVI, but the two are not the same object.
- Panel A's combined specification has a pseudo-R^2 of 13.00%, much larger than Panel B's 0.50%.

Economic message: Institutional attention is much more tightly linked to firm news and market-moving events than retail attention. Retail attention is noisier and less explained by the same covariates.

Table 3
Contemporaneous relation between abnormal institutional attention, abnormal retail attention, attention proxies, and other explanatory variables

A.* AIA *as a dependent variable

Variable	(1)	(2)	(3)	(4)	(5)	(6)
ANews t	0.400					0.374
	(86.16)					(77.66)
EarnAnnDum t	1.029					0.616
	(50.22)					(27.32)
RecChngDum t	1.116					0.767
	(62.31)					(43.68)
AbsDGTW t		0.063				0.041
		(20.23)				(7.36)
AVol t		0.055				0.034
		(6.09)				(6.19)
HLtoH t		3.196				11.275
		(10.01)				(14.48)
52 high dum t		0.364				−0.065
		(33.57)				(−5.57)
52 low dum t		0.100				−0.353
		(5.17)				(−13.77)
LnSize			0.153			0.226
			(26.20)			(35.48)
LnBM			0.019			0.032
			(1.97)			(3.19)
SDRET			0.037			−0.033
			(11.27)			(−5.64)
AdvExpToSale			0.192			0.090
			(1.20)			(0.54)
LnNumEst			0.286			0.268
			(20.87)			(18.28)
InstHold			−0.061			0.071
			(−2.19)			(2.21)
ADSVI t				0.176		0.081
				(13.37)		(8.48)
AEDGAR t				0.233		0.121
				(32.33)		(18.18)
Tuesday					−0.025	−0.077
					−1.58	(−3.71)
Wednesday					−0.056	−0.115
					−3.35	(−5.52)
Thursday					−0.057	−0.148
					−3.39	(−6.97)
Friday					−0.247	−0.312
					−14.97	(−14.07)

(a) Panel A: AIA as dependent variable.

Table 3
Continued

B.* DADSVI *as a dependent variable

Variable	(1)	(2)	(3)	(4)	(5)	(6)
ANews t	0.055					0.019
	(12.89)					(5.16)
EarnAnnDum t	0.303					0.164
	(11.95)					(7.00)
RecChngDum t	0.127					0.040
	(7.09)					(2.32)
AbsDGTW t		0.025				0.019
		(12.93)				(10.27)
AVol t		0.026				0.021
		(6.23)				(3.86)
HLtoH t		−0.283				0.756
		(−1.62)				(4.68)
52 high dum t		0.087				0.042
		(9.07)				(4.41)
52 low dum t		0.100				0.066
		(5.94)				(4.04)
LnSize			0.007			0.009
			(2.22)			(2.57)
LnBM			0.001			0.002
			(0.23)			(0.58)
SDRET			−0.014			−0.027
			(−5.87)			(−9.94)
AdvExpToSale			0.114			0.100
			(1.14)			(0.96)
LnNumEst			0.012			0.000
			(1.83)			(0.03)
InstHold			0.010			0.036
			(0.62)			(2.24)
AIA t				0.208		0.118
				(20.25)		(12.70)
AEDGAR t				0.028		0.011
				(8.06)		(3.56)
Tuesday					−0.011	−0.013
					−0.86	(−1.01)
Wednesday					−0.065	−0.066
					−4.74	(−4.87)
Thursday					−0.083	−0.089
					−5.76	(−6.24)
Friday					−0.137	−0.136
					−7.46	(−7.40)
P-RSQ	0.15%	0.21%	0.04%	0.20%	0.08%	0.50%

The table reports the results of the contemporaneous relation between our Abnormal Institutional Attention measure (*AIA*) from Bloomberg (panel A) and the abnormal retail attention dummy (*DADSVI*) based on Google's daily Search Volume Index (panel B) on selected explanatory variables. *AIA*, *DADSVI*, and other variables are defined in Table 1.
Panel A includes 3,144,109 day stock observations, and panel B includes 1,338,203 day stock observations. We handle *DADSVI*'s missing observations when analyzing *AIA* in panel A using Pontiff and Woodgate's (2008) approach. First, we define a dummy variable that takes a value of one whenever the *DADSVI* exists and zero otherwise. Then we replace *DADSVI* missing values with zeros. We repeat the same procedure for *AEDGAR*.
Each panel includes six identical specifications. For example in panel A, Specification 1 examines the relation between *AIA* and "News" variables; Specification 2 examines the relation between *AIA* and price related variables; Specification 3 examines the relation between *AIA* and other firm characteristics; Specification 4 examines the relation between *AIA* and other attention measures; Specification 5 examines the effect of the day of the week effect on *AIA*; and Specification 6 examines all five categories together. *P-RSQ* is the probit model's pseudo R-squared. Standard errors are clustered by firm and day and t-statistics are reported below the coefficient estimates.

(b) Panel B: DADSVI as dependent variable.

5.4 Table 4: Attention and abnormal trading volume

Read:

- The AIA coefficient is strongly positive for both abnormal CRSP volume and abnormal Ancerno institutional volume.
- In the full-control specification, the CRSP abnormal-volume difference is 0.237, and the Ancerno institutional abnormal-volume difference is 0.312.
- The difference-in-differences effect for AIA remains positive and statistically significant.
- Retail attention also predicts abnormal volume, but the institutional interpretation is weaker because the institutional-volume difference is not larger in the same way.

Economic message: AIA is not merely browsing. It is associated with actual trading, especially institutional trading.

Table 4
Abnormal institutional attention, abnormal retail attention, and abnormal trading volume

Variable	(1)	(2)	(3)	(4)	(5)	(6)
A. AIA and abnormal trading volume						
CRSP-AVol-diff	0.686	0.490	0.263	0.234	0.229	0.237
	(35.76)	(26.91)	(16.69)	(15.50)	(15.45)	(15.41)
Ancerno-AVol- diff	0.797	0.580	0.339	0.309	0.299	0.312
	(32.82)	(26.60)	(15.72)	(16.11)	(15.67)	(15.85)
Diff-In-diff	0.111	0.090	0.077	0.075	0.070	0.076
	(8.91)	(7.39)	(5.68)	(5.44)	(5.21)	(5.48)
Table 3 controls						
Control set 1		Yes	Yes	Yes	Yes	Yes
Control set 2			Yes	Yes	Yes	Yes
Control set 3				Yes	Yes	Yes
Control set 4					Yes	Yes
Control set 5						Yes
Adj-RSQ	0.61%	0.86%	2.27%	3.72%	3.74%	3.79%
B. DADSVI and Abnormal Trading Volume						
CRSP-AVol-diff	0.137	0.109	0.068	0.060	0.052	0.055
	(14.85)	(13.67)	(12.49)	(11.93)	(11.09)	(11.80)
Ancerno-AVol- diff	0.142	0.113	0.065	0.061	0.051	0.056
	(10.18)	(8.60)	(5.43)	(5.14)	(4.29)	(4.64)
Diff-In-diff	0.005	0.004	−0.003	0.000	−0.001	0.001
	(0.80)	(0.58)	(−0.30)	(0.04)	(−0.13)	(0.09)
Table 3 controls						
Control set 1		Yes	Yes	Yes	Yes	Yes
Control set 2			Yes	Yes	Yes	Yes
Control set 3				Yes	Yes	Yes
Control set 4					Yes	Yes
Control set 5						Yes
Adj-RSQ	0.06%	0.80%	3.09%	3.47%	3.49%	3.55%

The table reports the results of panel regressions of abnormal trading volume on abnormal institutional attention

Figure 5: Table 4: Abnormal institutional attention, abnormal retail attention, and abnormal trading volume.

5.5 Table 5 and Figure 3: Lead-lag relation among institutional attention, retail attention, and news

Read: Table 5:

- AIAC is highly persistent, but institutional attention also responds to lagged news.
- Lagged AIAC has a positive effect on ADSVI, while the effect of lagged ADSVI on AIAC is weak or negative.
- A one-standard-deviation shock to institutional attention predicts future retail attention.

Read: Figure 3:

- Panel A shows retail attention rises after an institutional-attention shock, while institutional attention does not rise after a retail-attention shock.
- Panel B shows institutional attention responds more strongly and more quickly to news shocks than retail attention.

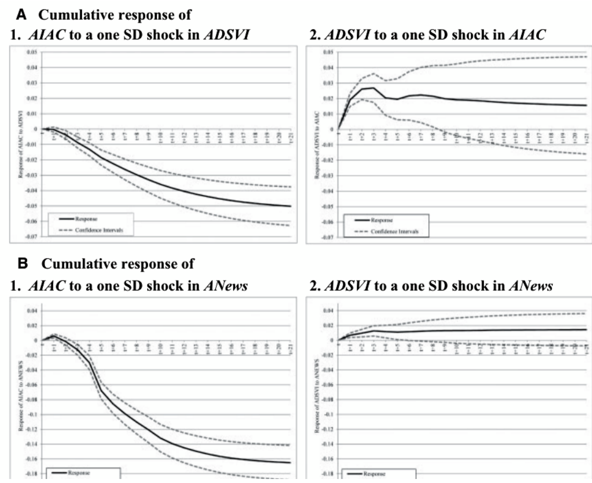
Economic message: Institutions appear to process news earlier. Retail investors follow with a lag.

Table 5
Lead-lag analysis of AIA, ADSVI and ANews

	AIAC t			*ADSVI t*		
Variable	(1)	(2)	(3)	(4)	(5)	(6)
AIAC t-1	0.276	0.245	0.245	0.028	0.019	0.019
	(80.34)	(72.10)	(72.30)	(11.94)	(8.90)	(8.90)
AIAC t-2	0.070	0.073	0.072	−0.003	−0.003	−0.003
	(27.57)	(29.63)	(29.64)	(−1.89)	(−1.70)	(−1.70)
AIAC t-3	0.057	0.058	0.058	−0.005	−0.005	−0.005
	(24.95)	(26.11)	(26.19)	(−2.89)	(−2.75)	(−2.75)
AIAC t-4	0.046	0.049	0.049	−0.009	−0.009	−0.009
	(19.88)	(21.37)	(21.29)	(−5.28)	(−4.99)	(−5.00)
AIAC t-5	0.075	0.076	0.076	0.000	0.000	0.000
	(28.19)	(28.87)	(28.91)	(0.14)	(0.26)	(0.26)
ADSVI t-1	0.003	0.000	0.000	0.260	0.259	0.259
	(2.13)	(−0.36)	(−0.37)	(53.94)	(53.77)	(53.77)
ADSVI t-2	−0.003	−0.003	−0.003	0.092	0.092	0.092
	(−4.22)	(−4.75)	(−4.66)	(36.31)	(36.26)	(36.26)
ADSVI t-3	−0.003	−0.003	−0.003	0.058	0.058	0.058
	(−3.84)	(−4.34)	(−4.27)	(27.37)	(27.26)	(27.25)
ADSVI t-4	−0.002	−0.001	−0.002	0.068	0.068	0.068
	(−2.08)	(−1.92)	(−1.98)	(24.37)	(24.41)	(24.42)
ADSVI t-5	−0.003	−0.002	−0.002	0.108	0.108	0.108
	(−3.35)	(−2.89)	(−2.78)	(28.23)	(28.25)	(28.25)
ANews t-1	0.015	0.006	0.006	0.010	0.007	0.007
	(5.64)	(4.37)	(4.31)	(5.75)	(4.32)	(4.32)
ANews t-2	−0.002	−0.010	−0.010	0.002	0.000	0.000
	(−1.82)	(−7.80)	(−7.86)	(1.14)	(−0.19)	(−0.20)
ANews t-3	−0.001	−0.008	−0.008	0.003	0.001	0.001
	(−0.57)	(−6.58)	(−6.67)	(2.32)	(0.95)	(0.95)
ANews t-4	−0.004	−0.012	−0.012	−0.001	−0.003	−0.003
	(−3.23)	(−10.22)	(−10.36)	(−0.46)	(−1.81)	(−1.81)
ANews t-5	−0.025	−0.029	−0.029	0.000	−0.001	−0.001
	(−18.12)	(−22.23)	(−22.55)	(0.10)	(−0.53)	(−0.55)
AVol t		0.046	0.046		0.016	0.016
		(8.64)	(8.63)		(6.80)	(6.81)
AbsRet t		0.049	0.049		0.017	0.017
		(16.70)	(16.69)		(13.45)	(13.45)
EarnAnnDum t		0.877	0.875		0.177	0.177
		(42.84)	(42.76)		(9.92)	(9.91)
RecChngDum t		0.593	0.593		0.044	0.044
		(37.51)	(37.48)		(3.66)	(3.67)
AveMktNews t			0.125			−0.039
			(3.14)			(1.89)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
AIAC-ADSVI diff						0.164
Wald test p-value						(0.01)
AdjRSQ	13.45%	19.03%	19.05%	15.96%	16.16%	16.16%

The table reports the results from panel vector autoregressions of *AIAC* and *ADSVI* on lagged *AIAC*, *ADSVI*, *ANews*, and other explanatory variables. The sample includes 1,338,203 day stock observations. See Tables 1 and 2 for variable and sample definitions. Because *ADSVI* and *ANews* are standardized, coefficients can be

(a) Table 5: VAR lead-lag coefficients.



(b) Figure 3: Cumulative impulse responses.

5.6 Table 6 and Figure 4: Institutional attention and post-earnings-announcement drift

Read: Table 6:

- The SUE coefficient is positive on the announcement day and remains positive for post-announcement drift horizons.
- The interaction SUE $\times$ AIA is positive on the announcement day, meaning attention strengthens immediate price response.
- The interaction becomes negative after the announcement and remains negative through long horizons.
- By $t+40$, the interaction is about as large as the baseline SUE drift coefficient in absolute value, implying little remaining drift when AIA $= 1$.

Read: Figure 4:

- For AIA $= 0$, positive earnings surprises drift upward and negative surprises drift downward.
- For AIA $= 1$, the drift is much smaller.
- The confidence bands show meaningful differences between high-attention and low-attention events.

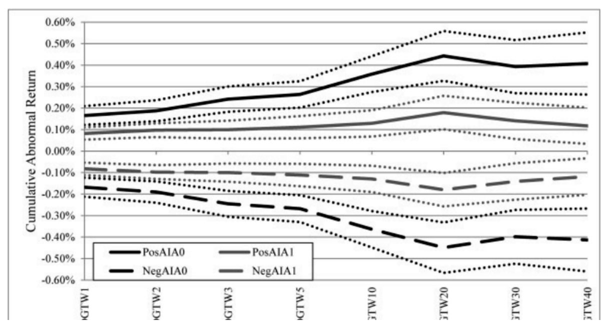
Economic message: PEAD is not universal. It is concentrated in earnings announcements that fail to attract institutional attention.

Table 6
Institutional attention and earnings announcements returns

A. Base case

	Day	Drift							
Variables	*t*	*t+1_t+1*	*t+1_t+2*	*t+1_t+3*	*t+1_t+5*	*t+1_t+10*	*t+1_t+20*	*t+1_t+30*	*t+1_t+40*
AIA t	0.0014	−0.0001	−0.0003	−0.0003	−0.0002	0.0000	0.0000	0.0017	0.0014
	(1.35)	(−0.21)	(−0.54)	(−0.40)	(−0.29)	(0.04)	(−0.00)	(0.96)	(0.75)
SUE t	0.0037	0.0006	0.0007	0.0009	0.0009	0.0013	0.0016	0.0014	0.0014
	(15.36)	(7.44)	(7.57)	(8.02)	(8.43)	(8.40)	(7.50)	(6.24)	(5.53)
SUE_AIA t	0.0008	−0.0003	−0.0003	−0.0005	−0.0005	−0.0008	−0.0009	−0.0009	−0.0010
	(2.69)	(2.64)	(−2.48)	(−4.00)	(−3.59)	(−4.30)	(−3.54)	(−3.03)	(−2.78)
ANews t	0.0018	0.0002	0.0003	0.0004	0.0005	0.0006	0.0008	0.0011	0.0016
	(4.37)	(1.11)	(1.19)	(1.51)	(1.37)	(1.44)	(1.53)	(1.54)	(2.04)
DADSVI t	0.0003	−0.0011	−0.0014	−0.0012	−0.0008	−0.0017	−0.0001	0.0006	0.0005
	(0.24)	(−1.96)	(−2.21)	(−1.66)	(−0.86)	(−1.35)	(−0.07)	(0.31)	(0.21)
AEDGAR t	−0.0011	0.0000	0.0001	−0.0002	−0.0001	0.0004	−0.0002	0.0002	−0.0012
	(−2.04)	(−0.04)	(0.38)	(−0.56)	(−0.11)	(0.76)	(−0.35)	(0.23)	(−1.18)
AVol t	−0.0012	0.0004	0.0004	0.0004	0.0005	0.0009	0.0011	0.0012	0.0011
	(−1.61)	(3.37)	(2.72)	(2.49)	(2.49)	(3.59)	(3.06)	(3.20)	(2.93)
HLtoH t	−0.1155	−0.0301	−0.0403	−0.0383	−0.0344	−0.0514	−0.0523	−0.0458	−0.0987
	(−3.29)	(−4.17)	(−3.72)	(−2.97)	(−2.03)	(−2.65)	(−1.97)	(−1.56)	(−2.97)
Ret t-5_t-1	−0.0007	0.0000	0.0000	0.0000	0.0000	−0.0001	−0.0004	−0.0004	0.0002
	(−5.32)	(0.32)	(0.77)	(0.41)	(−0.13)	(−0.75)	(−1.71)	(−1.38)	(0.67)
Turnover t-5_t-1	−0.1491	0.0061	0.0059	0.0617	0.0975	0.1175	−0.0394	−0.2679	−0.4344
	(−2.41)	(0.21)	(0.17)	(1.36)	(1.44)	(1.35)	(−0.34)	(−2.02)	(−2.75)
Spread t-5_t-1	−0.0704	−0.1926	−0.2600	−0.0487	−0.0373	−0.0641	0.2721	0.8670	1.2520
	(−0.15)	(−0.50)	(−0.57)	(−0.10)	(−0.06)	(−0.09)	(0.29)	(0.81)	(1.09)
SDRET	0.0011	0.0004	0.0004	0.0003	0.0000	0.0002	0.0008	0.0010	0.0037
	(1.96)	(1.65)	(1.27)	(0.59)	(0.09)	(0.25)	(0.60)	(0.67)	(1.35)
LnSize	−0.0027	−0.0001	−0.0002	−0.0004	−0.0007	−0.0014	−0.0010	−0.0011	−0.0007
	(−5.69)	(−0.50)	(−0.87)	(−1.20)	(−1.98)	(−2.86)	(−1.52)	(−1.34)	(−0.82)
LnBM	−0.0015	0.0001	0.0003	0.0004	0.0006	0.0003	0.0004	−0.0009	−0.0018
	(−2.75)	(0.22)	(0.74)	(1.14)	(1.10)	(0.39)	(0.43)	(−0.81)	(−1.37)
InstHold	0.0043	−0.0001	0.0000	0.0000	−0.0001	0.0000	0.0003	0.0004	0.0001
	(2.52)	(−0.17)	(−0.00)	(0.03)	(−0.09)	(0.01)	(0.12)	(0.12)	(0.02)
LnNumEst	0.0000	−0.0002	0.0002	0.0002	0.0002	0.0003	−0.0003	−0.0007	−0.0019
	(−0.03)	(−0.51)	(0.29)	(0.28)	(0.26)	(0.31)	(−0.18)	(−0.40)	(−0.94)
B. Adding interactions									
AIA t	0.0016	−0.0001	−0.0003	−0.0002	−0.0002	0.0001	0.0001	0.0016	0.0014
	(1.55)	(−0.14)	(−0.47)	(−0.33)	(−0.26)	(0.06)	(0.04)	(0.92)	(0.74)
DADSVI t	−0.0005	−0.0014	−0.0017	−0.0015	−0.0010	−0.0016	0.0002	0.0002	0.0003
	(−0.32)	(−2.40)	(−2.40)	(−1.95)	(−0.98)	(−1.25)	(0.13)	(0.10)	(0.12)
AEDGAR t	−0.0007	0.0000	0.0001	−0.0003	0.0000	0.0004	−0.0001	0.0001	−0.0011
	(−1.20)	(−0.10)	(0.38)	(−0.71)	(−0.09)	(0.68)	(−0.07)	(0.12)	(−0.95)
AVol t	−0.0014	0.0004	0.0004	0.0004	0.0005	0.0009	0.0011	0.0012	0.0011
	(−2.37)	(3.36)	(2.72)	(2.46)	(2.54)	(3.49)	(3.13)	(3.17)	(2.77)
ANEWS t	0.0011	0.0003	0.0003	0.0004	0.0005	0.0007	0.0008	0.0011	0.0016
	(2.94)	(1.40)	(1.26)	(1.62)	(1.47)	(1.62)	(1.32)	(1.45)	(1.90)
SUE t	0.0000	0.0006	0.0006	0.0008	0.0009	0.0012	0.0011	0.0011	0.0013
	(0.06)	(6.24)	(3.64)	(5.09)	(4.66)	(5.00)	(3.88)	(3.12)	(2.97)
SUE_AIA t	0.0002	−0.0003	−0.0003	−0.0005	−0.0005	−0.0008	−0.0008	−0.0008	−0.0010
	(0.76)	(−3.64)	(−2.73)	(−4.19)	(−3.58)	(−4.08)	(−3.09)	(−2.79)	(−2.85)
SUE_DADSVI t	0.0007	0.0003	0.0003	0.0003	0.0001	−0.0001	0.0000	0.0000	0.0002
	(1.42)	(1.83)	(1.31)	(1.62)	(0.54)	(−0.18)	(−0.02)	(−0.02)	(0.37)
SUE_AEDGAR t	−0.0001	0.0000	0.0000	0.0001	0.0000	0.0001	0.0000	−0.0001	−0.0002
	(−0.53)	(0.30)	(0.09)	(0.61)	(0.00)	(0.49)	(−0.22)	(−0.29)	(−0.70)
SUE_AVol t	0.0008	0.0000	0.0000	0.0000	0.0000	0.0001	0.0001	0.0001	0.0001
	(5.78)	(0.81)	(0.82)	(0.99)	(1.26)	(1.47)	(1.80)	(1.38)	(1.32)
SUE_ANewst	0.0007	−0.0001	0.0000	0.0000	−0.0001	−0.0001	−0.0001	−0.0001	0.0000
	(5.13)	(−1.41)	(−0.21)	(−0.63)	(−0.68)	(−1.16)	(−0.41)	(−0.41)	(−0.15)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
DADSVI-AIA Interaction diff	0.00048	0.00058	0.00060	0.00085	0.00067	0.00076	0.00079	0.00081	0.00117
Wald test *p*-value	0.37	0.00	0.01	0.00	0.02	0.03	0.06	0.07	0.05

(a) Table 6, panels A–B: earnings returns.



(b) Figure 4: Earnings drift by attention.

Table 6
Continued

C. Earnings announcements after market close

	Day	Drift							
Variables	*t*	*t+1_t+1*	*t+1_t+2*	*t+1_t+3*	*t+1_t+5*	*t+1_t+10*	*t+1_t+20*	*t+1_t+30*	*t+1_t+40*
AIA t-1	0.0008	−0.0002	0.0000	0.0000	0.0008	0.0005	0.0026	0.0031	0.0010
	(0.53)	(−0.44)	(−0.04)	(−0.06)	(0.78)	(0.34)	(1.34)	(1.27)	(0.36)
SUE t-1	0.0044	0.0006	0.0007	0.0008	0.0009	0.0012	0.0015	0.0010	0.0008
	(10.27)	(5.71)	(5.36)	(5.53)	(4.69)	(4.51)	(4.95)	(2.39)	(1.57)
SUE t -1_AIA t-1	0.0001	−0.0004	−0.0005	−0.0006	−0.0006	−0.0010	−0.0009	−0.0006	−0.0004
	(0.14)	(−3.29)	(−2.80)	(−2.91)	(−2.35)	(−3.04)	(−2.33)	(−1.89)	(−1.14)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

The table reports the results of panel regressions of earnings announcements' day-t and cumulative day *t*+1 to *t*+40 *DGTW* risk-adjusted returns on abnormal institutional attention and other explanatory variables. The sample includes 34,440 firm-quarter observations (see Tables 1 and 2 for variable and sample definitions). In panel A, we present our base case. In panel B, we explore the robustness of our results by adding additional interaction variables. In panel C, we focus on a reduced sample of earnings announcements that occur from 4:00 p.m.-11:59 p.m. of day *t*-1 (after market close). We use I/B/E/S timestamps, which are reasonable for this

Figure 8: Table 6, panel C: Earnings announcements after market close.

5.7 Table 7 and Figure 5: Institutional attention and recommendation-change drift

Read: **Table 7:**

- RecChng has a strong announcement-day return effect.
- The interaction RecChng × AIA is positive on day 0, showing stronger immediate reaction when institutions pay attention.
- The interaction is negative over future horizons, meaning drift is reduced for high-attention recommendation changes.
- The same pattern remains after including other attention interactions.

Read: Figure 5:

- Recommendation-change drift is strongest for AIA = 0 events.
- Positive recommendation changes with AIA = 0 drift upward; negative changes with AIA = 0 drift downward.
- AIA = 1 events show much less drift.

Economic message: The attention mechanism is not specific to earnings. It also explains slow price adjustment to analyst recommendation changes.

Table 7
Institutional attention and change-in-analyst-recommendations returns

A. Base case

	Day 0	Drift									
Variables	*t*	*t+1_t+1*	*t+1_t+2*	*t+1_t+3*	*t+1_t+4*	*t+1_t+5*	*t+1_t+6*	*t+1_t+7*	*t+1_t+8*	*t+1_t+9*	*t+1_t+*
AIA t	0.0002	0.0001	0.0001	0.0001	0.0000	−0.0002	0.0006	0.0001	0.0000	0.0002	−0.000
	(0.30)	(0.31)	(0.20)	(0.08)	(0.04)	(−0.20)	(0.69)	(0.15)	(0.02)	(0.19)	(−0.23)
RecChng t	0.0088	0.0015	0.0018	0.0019	0.0017	0.0018	0.0021	0.0022	0.0021	0.0025	0.002
	(33.33)	(8.15)	(7.86)	(6.42)	(4.90)	(4.89)	(5.16)	(5.20)	(4.61)	(5.22)	(4.86)
RecChng_AIA t	0.0083	−0.0007	−0.0009	−0.0008	−0.0005	−0.0006	−0.0010	−0.0012	−0.0012	−0.0018	−0.002
	(16.61)	(−2.71)	(−2.43)	(−1.94)	(−1.49)	(−1.53)	(−1.83)	(−2.00)	(−1.78)	(−2.64)	(−2.75)
ANews t	0.0000	−0.0001	0.0001	0.0003	0.0001	0.0000	−0.0001	0.0003	0.0002	0.0004	0.000
	(0.04)	(−0.53)	(0.23)	(0.82)	(0.28)	(0.00)	(−0.20)	(0.55)	(0.41)	(0.65)	(0.67)
DADSVI t	0.0022	0.0001	0.0004	0.0004	0.0010	0.0011	0.0010	0.0015	0.0008	0.0013	0.001
	(1.36)	(0.08)	(0.34)	(0.36)	(0.68)	(0.74)	(0.64)	(0.87)	(0.41)	(0.66)	(0.71)
AEDGAR t	0.0005	−0.0002	−0.0004	−0.0001	−0.0004	−0.0005	−0.0006	0.0002	0.0005	0.0004	0.000
	(0.89)	(−0.70)	(−0.97)	(−0.19)	(−0.78)	(−0.82)	(−0.91)	(0.25)	(0.66)	(0.62)	(0.80)
AVol t	0.0013	0.0006	0.0002	0.0003	0.0003	0.0004	0.0003	0.0003	0.0003	0.0003	0.000
	(1.55)	(2.00)	(1.18)	(1.52)	(1.75)	(2.09)	(1.18)	(1.08)	(1.01)	(1.11)	(1.59)
HLtoH t	−0.1921	−0.0368	−0.0252	−0.0371	0.0049	−0.0128	−0.0280	−0.0296	−0.0289	−0.0358	−0.048
	(−3.27)	(−1.66)	(−0.76)	(−1.00)	(0.11)	(−0.32)	(−0.66)	(−0.68)	(−0.59)	(−0.75)	(−1.09)
Ret t-5_t-1	0.0001	0.0000	0.0000	0.0000	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.000
	(1.23)	(0.16)	(0.57)	(0.24)	(0.55)	(0.61)	(0.93)	(0.98)	(0.83)	(0.71)	(0.23)
Turnover t-5_t-1	−0.0697	0.0118	−0.0050	−0.0228	−0.0472	−0.0409	−0.0422	−0.0304	−0.0635	−0.0658	−0.069
	(−1.97)	(0.70)	(−0.18)	(−0.93)	(−1.69)	(−1.35)	(−1.30)	(−0.84)	(−1.69)	(−1.72)	(−1.78)
Spread t-5_t-1	−1.4321	0.1499	−0.1295	−0.0527	0.1855	−0.5804	−1.8651	−2.3099	−2.7174	−2.4451	−3.124
	(−0.76)	(0.32)	(−0.20)	(−0.06)	(0.14)	(−0.37)	(−0.76)	(−1.04)	(−1.20)	(−0.96)	(−1.36)
SDRET	0.0005	0.0002	0.0004	0.0004	0.0004	0.0003	0.0000	0.0000	0.0001	−0.0001	−0.000
	(1.07)	(0.87)	(1.36)	(0.98)	(0.70)	(0.48)	(−0.03)	(0.04)	(0.09)	(−0.09)	(−0.14)
LnSize	−0.0021	−0.0003	−0.0002	−0.0001	0.0000	−0.0001	−0.0005	−0.0001	−0.0004	−0.0003	−0.000
	(−4.07)	(−1.48)	(−0.80)	(−0.46)	(0.03)	(−0.18)	(−0.94)	(−0.27)	(−0.65)	(−0.45)	(−0.66)
LnBM	0.0003	0.0004	0.0007	0.0003	−0.0001	−0.0002	−0.0004	−0.0007	−0.0004	−0.0006	−0.000
	(0.53)	(1.59)	(2.02)	(0.79)	(−0.22)	(−0.31)	(−0.65)	(−1.14)	(−0.57)	(−0.78)	(−0.54)
InstHold	−0.0001	0.0013	−0.0003	0.0005	0.0034	0.0039	0.0050	0.0051	0.0051	0.0053	0.005
	(−0.03)	(1.34)	(−0.22)	(0.31)	(1.83)	(1.91)	(2.11)	(2.03)	(1.93)	(1.91)	(1.88)
LnNumEst	0.0006	0.0005	0.0003	0.0004	0.0005	−0.0001	0.0000	−0.0007	−0.0002	−0.0004	−0.000
	(0.59)	(1.22)	(0.47)	(0.49)	(0.56)	(−0.13)	(0.00)	(−0.61)	(−0.13)	(−0.28)	(−0.65)

Figure 9: Table 7, panel A: Recommendation-change returns, base case.

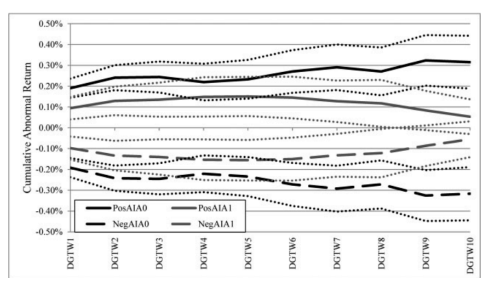
Table 7
Continued

B. Adding interactions

	Day 0	Drift									
Variables	*t*	*t+1_t+1*	*t+1_t+2*	*t+1_t+3*	*t+1_t+4*	*t+1_t+5*	*t+1_t+6*	*t+1_t+7*	*t+1_t+8*	*t+1_t+9*	*t+1_t+*
AIA t	0.0001	0.0001	0.0001	0.0001	0.0000	−0.0002	0.0006	0.0001	0.0000	0.0002	−0.000
	(0.19)	(0.29)	(0.19)	(0.09)	(0.03)	(−0.19)	(0.68)	(0.13)	(0.02)	(0.17)	(−0.20)
DADSVI t	0.0017	0.0002	0.0005	0.0007	0.0012	0.0013	0.0012	0.0017	0.0009	0.0014	0.001
	(1.16)	(0.18)	(0.45)	(0.55)	(0.85)	(0.82)	(0.77)	(0.97)	(0.52)	(0.74)	(0.76)
AEDGAR t	0.0002	−0.0003	−0.0004	−0.0001	−0.0004	−0.0005	−0.0006	0.0001	0.0004	0.0004	0.000
	(0.39)	(−0.66)	(−1.01)	(−0.25)	(−0.81)	(−0.83)	(−0.94)	(0.20)	(0.62)	(0.59)	(0.75)
AVol t	0.0025	0.0007	0.0003	0.0004	0.0004	0.0005	0.0004	0.0004	0.0004	0.0004	0.000
	(2.45)	(2.18)	(1.62)	(2.05)	(1.90)	(2.41)	(1.47)	(1.37)	(1.28)	(1.36)	(2.06)
ANews t	−0.0004	−0.0001	0.0001	0.0003	0.0001	0.0000	−0.0001	0.0003	0.0002	0.0004	0.000
	(−0.73)	(−0.61)	(0.23)	(0.83)	(0.32)	(0.05)	(−0.19)	(0.53)	(0.41)	(0.66)	(0.69)
RecChng t	0.0039	0.0010	0.0015	0.0015	0.0015	0.0014	0.0017	0.0018	0.0015	0.0020	0.001
	(3.99)	(2.70)	(4.98)	(4.11)	(3.75)	(3.41)	(3.68)	(3.76)	(2.95)	(3.60)	(3.14)
RecChng_AIA t	0.0035	−0.0008	−0.0009	−0.0009	−0.0005	−0.0006	−0.0011	−0.0015	−0.0013	−0.0019	−0.002
	(7.11)	(−2.97)	(−2.52)	(−2.01)	(−1.45)	(−1.49)	(−1.83)	(−2.20)	(−1.79)	(−2.48)	(−2.64)
RecChng_DADSVI t	0.0019	−0.0003	−0.0003	−0.0004	−0.0004	−0.0004	−0.0003	0.0000	−0.0001	0.0002	0.000
	(1.61)	(−0.40)	(−0.37)	(−0.87)	(−0.58)	(−0.34)	(−0.64)	(−0.03)	(0.29)	(0.15)	(0.05)
RecChng_AEDGAR t	0.0012	0.0000	0.0001	0.0002	0.0002	0.0001	0.0001	−0.0001	−0.0002	−0.0002	−0.000
	(2.93)	(−0.07)	(0.24)	(0.68)	(0.51)	(0.17)	(0.17)	(−0.19)	(−0.36)	(−0.44)	(−0.37)
RecChng_AVol t	0.0026	0.0003	0.0002	0.0003	0.0002	0.0002	0.0003	0.0003	0.0003	0.0003	0.000
	(4.05)	(1.62)	(2.17)	(2.46)	(1.40)	(1.61)	(1.64)	(1.67)	(1.79)	(1.63)	(2.40)
RecChng_Anews t	0.0051	−0.0002	−0.0001	−0.0001	−0.0003	−0.0001	0.0000	0.0001	0.0000	−0.0001	−0.000
	(10.88)	(−0.77)	(−0.34)	(−0.57)	(−1.17)	(−0.37)	(0.15)	(0.41)	(−0.10)	(−0.31)	(−0.24)
Controls	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
DADSVI-AIA Interaction diff	−0.00166	0.00050	0.00063	0.00054	0.00004	0.00016	0.00080	0.00142	0.00125	0.00212	0.002
Wald test *p*-value	0.19	0.39	0.41	0.35	0.88	0.80	0.32	0.21	0.30	0.11	0.07

The table reports the results of panel regressions of change in analyst recommendations' day-*t* and cumulative day *t*+1 to *t*+10 *DGTW* risk-adjusted returns on institutional attenti and other explanatory variables. In constructing the sample, we follow Jegadeesh and Kim (2010), Loh and Stulz (2011), and Kadan, Michaely, and Moulton (2013) to identify releva changes in analyst recommendations. In particular, we (1) remove recommendation changes that occur on the same day as, or the day following, earnings announcements; (2) remo recommendation changes on days when multiple analysts issue recommendations for the same firm; (3) require at least one analyst to have issued a recommendation for the stock a revised the recommendation within 180 calendar days; (4) require at least two analysts to have active recommendations for the stock as of the day before the revision; and (5) conside recommendation to be active for up to 180 days after it is issued or until I/B/E/S indicates that the analyst has stopped issuing recommendations for that stock. After applying all of the filters, we obtain 16,312 changes in recommendations.
In panel A, we present our base case. In panel B, we explore the robustness of our results by adding additional interaction variables. *RecChng* is the change in analyst recommendations. T variable ranges from −4 to 4, where a positive (negative) number refers to an upgrade (a downgrade). *RecChng_AIA* is the interaction between *RecChng* and *AIA*. Similar to Table 6, sin *AIA* (*DADSVI*) is a dummy variable, the interaction with *RecChng* measures the additional sensitivity of the *AIA*=1 (*DADSVI*=1) group. In panel 7.B, *DASVI-AIA interaction diff* is t

(a) Table 7, panel B: additional interactions.



(b) Figure 5: recommendation drift by attention.

5.8 Table 8: Calendar-time trading strategies

Read:

- For earnings announcements, the AIA = 0 long-short strategy earns 0.95% over five days and 1.07% over ten days using raw returns.
- The AIA = 1 earnings strategy earns much less: 0.29% over five days and 0.16% over ten days.
- The difference portfolio is economically large and statistically significant.
- Analyst recommendation strategies show the same pattern: AIA = 0 returns are high, AIA = 1 returns are close to zero, and the difference is meaningful.

Economic message: The drift documented in the regressions is tradable in calendar time mainly among low institutional-attention events.

Table 8
Earnings announcements and analyst recommendation changes calendar time portfolios

A. Earnings announcements

	5 trading days			10 trading days		
	RET	FF3	FF5	RET	FF3	FF5
LS AIA=0	0.95%	0.92%	0.91%	1.07%	1.03%	1.03%
	(5.45)	(5.26)	(5.18)	(4.44)	(4.26)	(4.28)
LS AIA=1	0.29%	0.23%	0.23%	0.16%	0.07%	0.07%
	(1.70)	(1.62)	(1.59)	(0.81)	(0.32)	(0.37)
DIFF	0.65%	0.68%	0.67%	0.91%	0.96%	0.96%
	(3.20)	(3.31)	(3.26)	(2.44)	(2.55)	(2.52)
B. Analyst recommendation changes						
LS AIA=0	0.64%	0.63%	0.63%	0.66%	0.62%	0.61%
	(4.08)	(4.00)	(3.96)	(3.35)	(3.17)	(3.14)
LS AIA=1	0.19%	0.17%	0.17%	0.05%	0.04%	0.04%
	(1.76)	(1.67)	(1.66)	(0.41)	(0.23)	(0.21)
DIFF	0.45%	0.46%	0.46%	0.60%	0.58%	0.57%
	(1.65)	(1.88)	(1.85)	(2.17)	(2.36)	(2.31)

The table reports the results from calendar time portfolios of earnings announcements (panel A) and changes in analyst recommendation (panel B) strategies for portfolios that are held for five and ten trading days. See Tables 6 and 7 for event and sample definitions. To construct the earnings announcements strategies, we use these portfolios to construct the following trading strategies: (1) [AIA=0, SUE>0 minus AIA=0, SUE<0], labeled

Figure 11: Table 8: Earnings announcements and analyst recommendation calendar-time portfolios.

6 Short summary

- The paper introduces AIA, a direct stock-day measure of abnormal institutional attention based on Bloomberg terminal news reading and search activity.
- AIA = 1 when Bloomberg's abnormal attention score is 3 or 4.
- Bloomberg terminal users are primarily professionals in banking, asset management, institutional financial services, and related roles.
- Retail attention is measured separately with Google search activity.
- AIA is related to news, earnings announcements, recommendation changes, volatility, volume, EDGAR search, and retail search, but it is not reducible to any of them.
- Institutional attention is more strongly connected to institutional trading volume than retail attention is.
- Institutional attention responds quickly to news and leads retail attention.
- Institutional attention strengthens immediate price responses to earnings news and analyst recommendation changes.
- Institutional attention reduces subsequent price drift after those events.
- PEAD and recommendation drift are concentrated in events with AIA = 0.
- Long-short trading strategies earn larger returns among low-attention events than high-attention events.
- The broad lesson is that attention only matters for price efficiency when it comes from investors who can rapidly process and trade on the information.

7 Conclusion

7.1 Core conclusion

Ben-Rephael, Da, and Israelsen show that institutional attention is a key input into the price-discovery process. Their direct Bloomberg-based measure reveals that institutional investors often pay attention before retail investors and that their attention helps prices incorporate public information faster.

7.2 Economic intuition

Public information does not automatically move prices to efficient levels. Someone must notice it, interpret it, and trade. Institutions are more likely to have the resources, incentives, and capital to do this quickly. When institutions pay attention, prices adjust immediately. When they fail to pay attention, the same type of news produces delayed adjustment and predictable drift.

7.3 Takeaway

The paper refines the limited-attention literature by showing that investor type matters. Retail attention may be visible and useful, but institutional attention is more important for permanent price adjustment. The empirical pattern is simple and powerful: underreaction survives when institutions are inattentive.